

Project Proposal

Team: Insightfy

Data

Adventures works

Tools

SQL-Python-Power BI

Team

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- Data Cleaning
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- Data modeling
- Visualization idea
- Dashboard Design
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Project Description

This project analyzes the performance of sales operations and supply chain activities using the AdventureWorks dataset. The objective is to evaluate how product demand, inventory levels, and supplier performance impact overall sales efficiency and business operations.

The analysis focuses on identifying sales trends, top-performing products, customer purchasing patterns, and potential supply chain inefficiencies such as low inventory levels or delayed shipments. By integrating sales data with product and inventory information, the project aims to provide insights that support better demand planning, inventory optimization, and strategic decision-making.

The final deliverables include an interactive dashboard that highlights key business metrics, enabling stakeholders to monitor performance and identify opportunities for improving operational efficiency and profitability.

KPIs

Sales KPIs

1. Total Revenue

Total sales generated from all completed orders.

2. Total Orders

Total number of customer orders placed.

3. Total Units Sold

Total quantity of products sold across all orders.

4. Average Order Value (AOV)

Average revenue generated per order.

Formula:

$\text{Total Revenue} / \text{Total Orders}$

5. Revenue by Product Category

Total revenue generated by each product category.

6. Top Selling Products

Products generating the highest sales volume or revenue.

7. Monthly Sales Growth Rate

Percentage change in revenue compared to the previous month.

Customer KPIs

8. Total Customers

Total number of unique customers who placed orders.

9. Revenue per Customer

Average revenue generated per customer.

10. Top Customers by Revenue

Customers contributing the highest revenue.

Supply Chain KPIs

11. Inventory Level

Current quantity available for each product in inventory.

12. Low Stock Products

Number of products where inventory is below the reorder point.

13. Inventory Turnover Rate

How quickly inventory is sold and replaced over a period.

14. Supplier Order Volume

Total number of purchase orders placed with each supplier.

15. Total Purchasing Cost

Total cost spent on purchasing products from suppliers.

Operational KPIs

16. Order Fulfillment Time

Average time between order placement and shipment.

Formula:

$\text{ShipDate} - \text{OrderDate}$

17. On-Time Delivery Rate

Percentage of orders shipped on or before the due date.

18. Average Shipping Delay

Average delay between expected shipping date and actual shipping date